

ASSIGNMENT

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

Native development creates an app specifically for one operating system, such as Android or iOS. It usually provides better performance and access to device features, so it is commonly used for apps that need high performance or advanced platform features.

Hybrid development uses web technologies such as HTML, CSS, and JavaScript inside a native wrapper. It is often used for simpler applications because one codebase can support multiple platforms.

Cross-platform development allows developers to create apps for different platforms using a shared codebase. It is commonly used to reduce development time and cost. Flutter is an example of a cross-platform framework.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

The **Flutter Ecosystem** includes Flutter's framework, Dart programming language, widgets, packages, libraries, and development tools. These resources help developers build user interfaces and add features to applications.

Flutter also allows developers to use one codebase for applications on multiple platforms. This makes development easier and can reduce the time needed to create and maintain applications.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

The Software Development Life Cycle (SDLC) is a structured process for developing and maintaining software. Its common phases are planning, analysis, design, coding, testing, deployment, and maintenance.

SDLC is important because it gives developers an organized process to follow. It helps teams understand requirements, manage resources, find problems, and create software that meets user needs.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Flutter and SDLC work together by combining a development technology with a development process. Developers can use SDLC to plan and analyze the application, design its interface, and then use Flutter and Dart during coding. After that, they test, deploy, and maintain the application.

This approach helps developers stay organized while using Flutter's ability to build applications for multiple platforms from a shared codebase

5. Why should developers understand these concepts before starting an application project?

Developers should understand these concepts because they help them choose the right development approach and plan the project properly. Knowing the differences between Native, Hybrid, and Cross-platform development helps them choose suitable technology, while understanding Flutter and SDLC helps them develop, test, and maintain the application more effectively.